CSCI4146/6409 Process of Data Science
Project 4: Responsible AI Fairness Monitoring in Production

This project is designed as a responsible-AI model-risk engagement, not a leaderboard exercise. You are hired as an external audit team for a mortgage decision-support system trained on historical HMDA loan application records. Your goal is to produce evidence about model quality, fairness risks, mitigation trade-offs, and production monitoring controls.

Project logistics:

- Programming language: Python (Jupyter Notebook environment)
- Primary dataset: prepared HMDA slice in `project 4/data/hmda_project4_slice.csv.gz`
- Workbook: `project_workbook.ipynb`
- Due date: posted in Brightspace/course syllabus
- Team policy: follow current course policy (individual or team-based as announced)

Marking scheme and requirements: Full marks are given for (1) working, readable, reasonably efficient, documented analysis code, and (2) responsible-AI reasoning supported by quantitative evidence. High model accuracy alone is insufficient. Fairness claims must include sample sizes, metric definitions, and limitations.

Please follow course collaboration and integrity policy. If you use external resources, including AI tools, you must disclose them and remain fully responsible for correctness, interpretation, and final arguments.

What/how to submit your work:

- Completed `project_workbook.ipynb` with outputs visible.
- PDF export of the notebook.
- Concise executive/model-risk memo (1–2 pages, separate PDF or clearly labeled notebook section).
- Model card, either as a clearly labeled notebook section or exported artifact.
- Source and AI-usage disclosure, if applicable.

1 Before You Start

In the tutorial notebook, you practiced a compact fairness workflow on a credit-risk style dataset. In this project, you must apply the same ideas to a real, substantial HMDA mortgage dataset.

Frequently asked clarifications

- *Do we need the highest possible AUC?* No. You need a credible model and a defensible responsible-AI audit.
- *Can we use one fairness metric only?* No. You must compare multiple fairness definitions and explain trade-offs.
- *Can we remove protected attributes and declare the model fair?* No. Protected attributes may still be reconstructed through proxies.
- *Do alerts prove discrimination?* No. Alerts initiate investigation; they do not automatically establish unlawful discrimination.
- *Can small groups drive strong claims?* Be careful. You must report sample sizes and treat unstable small-cell estimates cautiously.

2 Dataset

The provided slice is built from the official CFPB/FFIEC HMDA Data Browser API. It contains California HMDA records for 2022–2024, filtered to home-purchase applications with action taken in originated, approved but not accepted, or denied. Local post-filters retain first-lien and principal-residence records when those fields are present.

The target is `target_denied`, where 1 indicates historical denial and 0 indicates historical origination or approval. This target is a historical institutional action, not objective creditworthiness. You must discuss label limitations, missing information, historical bias, selection effects, and human discretion. Withdrawn, incomplete, purchased, preapproval, and otherwise ambiguous records are not part of the prepared modeling slice. You must acknowledge this exclusion and explain why those records should not be treated as either approved or denied.

3 Main Assignment

Complete all required tasks in `project_workbook.ipynb`. The milestone structure below is mandatory.

M0. Data and Model Risk Assessment (15%)

Minimum requirements:

- Load the HMDA slice and inspect schema, row count, target distribution, and missingness.
- Define the model-assisted decision and affected population.
- Document outcome mapping and acknowledge ambiguous or excluded records that are absent from the prepared slice.

- Identify protected and vulnerable groups.
- Report protected-group sample sizes and missing demographic values.
- Explain why historical action outcomes are not neutral ground truth.

M1. Baseline Models and Threshold Policy (15%)

Minimum requirements:

- Build leakage-safe features available at decision time.
- Use train/validation/test discipline; use a time-aware split when appropriate.
- Train at least two models: one interpretable baseline and one higher-capacity model.
- Report ROC-AUC or PR-AUC, precision, recall, F1, confusion matrix, and calibration evidence.
- Select an operational threshold on validation data only.
- Explain the threshold's utility, capacity, and error trade-offs.

M2. Fairness Audit Across Definitions (25%)

Minimum requirements:

- Analyze sex, race/ethnicity, age, and at least two intersectional groups.
- Include sample sizes in all fairness tables.
- Report at least three fairness definition families, such as demographic parity, equal opportunity, equalized odds, predictive parity, calibration within groups, and worst-group performance.
- Identify where fairness definitions disagree.
- Define and compute an individual fairness diagnostic.
- Perform proxy analysis by attempting to reconstruct at least one protected attribute from non-protected features.
- Explain which fairness definition receives the most weight in your deployment recommendation and why.

M3. Mitigation and Trade-Off Analysis (20%)

Implement at least two mitigation approaches from different categories.

- Data or feature intervention: reweighting, resampling, proxy feature removal, or feature transformation.
- Training intervention: fairness-aware objective, group-robust learning, or cost-sensitive learning.
- Decision intervention: threshold adjustment, reject-option review, calibrated post-processing, or human-review routing.

You must compare baseline and mitigated systems using utility, calibration, and multiple fairness metrics. State which groups benefit, which groups remain at risk, and what trade-offs are created.

M4. Production Monitoring and Incident Response (15%)

Minimum requirements:

- Create production-era batches using time, geography, lender, or another justified ordering.
- Track data quality, feature/score drift, protected-group composition, performance by group, and fairness by group/intersection.
- Define alert thresholds and minimum sample-size rules.
- Trigger at least one simulated alert.
- Complete an incident investigation worksheet with causes, containment, remediation, and return-to-normal conditions.

M5. Final Recommendation and Model Card (10%)

Your final deployment decision must be one of:

- deploy as-is;
- deploy with controls;
- return for remediation;
- reject for this use case.

Link your decision to model, fairness, mitigation, and monitoring evidence. Include a model card documenting intended use, out-of-scope use, audited groups, limitations, monitoring commitments, and human oversight.

4 Required Artifacts

- Completed notebook with visible outputs.
- Exported artifacts under `artifacts/`.
- Fairness-definition decision log.
- Mitigation trade-off log.
- Incident investigation worksheet.
- Final evidence reference map.
- Model card and final deployment recommendation.

5 Marking Scheme

Component	Weight
M0 Data and model-risk assessment	15%
M1 Predictive modeling and threshold policy	15%
M2 Fairness audit across definitions	25%
M3 Mitigation and trade-off analysis	20%
M4 Monitoring and incident response	15%
M5 Communication, model card, and final recommendation	10%

6 AI/LLM Usage and Integrity

AI tools may be used as support tools, but not as replacements for your reasoning. If used, you must disclose usage and verify all generated content. Any submission that cannot be explained by the student(s) may be penalized under course integrity policy.

7 Final Submission Checklist

Before submission, ensure:

- Notebook runs end-to-end with visible outputs.
- M0–M5 requirements are all addressed.
- Target-label limitations and historical-bias risks are discussed.
- Threshold is chosen on validation data only.
- Fairness tables include sample sizes.
- At least three fairness definition families are compared.
- Individual fairness and proxy analyses are included.
- Two mitigation categories are compared.
- Monitoring alert and incident investigation are complete.
- Model card and evidence reference map are complete.
- Source and AI-use disclosures are included, if applicable.

End Note

The educational goal of this project is to practice responsible data science as production governance: define the decision, audit harms, compare mitigation trade-offs, monitor deployment, and communicate limitations honestly.